

MSC DISSERTATION · RESEARCH RESULTS

Examining and Comparing the Algorithms of the Adaptive Chess Puzzle Advisor

Measurement validity, a pre-registered replication, a paired simulation study, prequential calibration on real logs, and a real-player cohort evaluation, all generated from the result files in `eval/research/`.

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1 Measurement validity: the tactic labeller

Every weakness estimate in the system depends on the tactic label attached to each missed move. It decides which bandit arm the evidence goes to. So I measured the original labeller (`generator._classify_tactic`) against an independent reference: the themes of Lichess puzzles, resolved with the same priority rule the puzzle pool uses. Lichess themes come from an automatic tagger too, so these numbers measure *agreement*, not accuracy against human judgement. There are two samples: a uniform one of 30,000 puzzles (for prevalence-true agreement and κ) and a stratified one of 48,000 puzzles, up to 2,000 per class (for per-class recall).

The original labeller scored $\kappa = 0.23$ ("fair" agreement on the Landis & Koch scale [8]). It could never emit 8 of the 23 categories. Its failures were structural. It looked at a single move, so a mate in three was labelled as its opening check. It treated the king as a high-value fork target. And its skewer and X-ray rules were geometrically wrong. The replacement, `src/puzzles/tactic_tagger.py`, tags the engine's whole principal variation, uses the engine's mate verdict, gets pin and skewer from ray geometry, and takes endgame type from the material left on the board. It was developed on one sample (seed 7) and then scored on a **fresh held-out sample** (seed 11):

Metric	Old labeller	New tagger (held-out)	New tagger (development)
Strict agreement	31.2%	58.4%	57.6%
Lenient agreement (any mapped theme)	37.4%	60.9%	60.2%
Cohen's κ	0.23	0.52	0.51
Macro recall (stratified)	14.4%	32.0%	31.6%
Macro precision (uniform)	30.3%	53.7%	53.4%
Categories never emitted	8	5	—

Category	Recall old	Recall new	Precision old	Precision new
Mating Pattern	0.46	1.00	1.00	1.00
Fork	0.82	0.96	0.25	0.61
Skewer	0.12	0.76	0.06	0.36
Discovered Attack	0.08	0.69	0.71	0.58
Promotion	0.08	0.64	0.77	0.87
Pin	0.20	0.23	0.42	0.43
Hanging Piece	0.59	0.57	0.33	0.53

Tactic labeller vs Lichess puzzle themes (row-normalised, stratified sample)

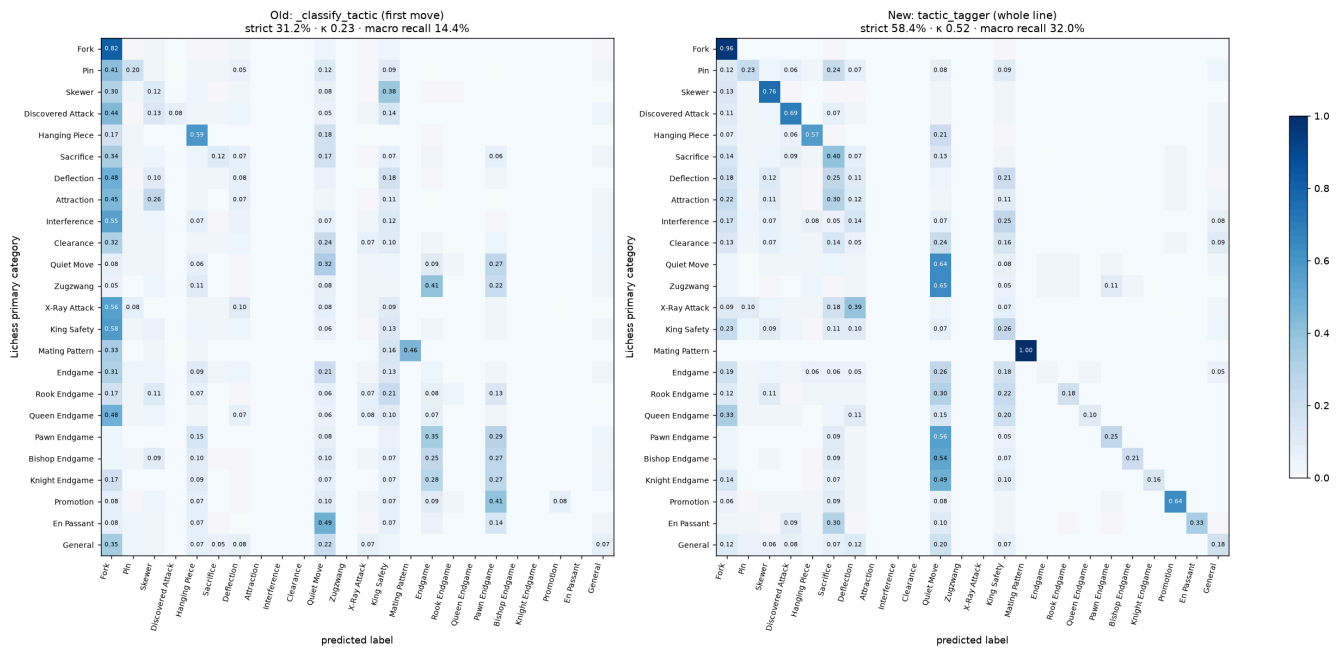


Figure 1. Confusion matrices, old and new, on the held-out sample, normalised by row.

What still cannot be detected

Attraction, Interference, Clearance, Zugzwang, X-Ray Attack: these need deeper reasoning about the line. The new tagger deliberately emits nothing for them rather than guessing. The old X-Ray rule's guesses had a precision of 0.

2 Simulation study

Synthetic players with known weaknesses, puzzle difficulties drawn from the real Lichess distribution for each category, game evidence mislabelled with the measured confusion matrix of the real tagger, and the production policy classes run unmodified. Every comparison is paired: 200 seeds, each facing every policy with the same player. Tests are paired t-tests with Holm correction [9], effect sizes are Cohen's d_z , and CIs are bootstrap.

E1 · Pre-registered replication

This is the protocol fixed before any result existed: three categories solved 30% of the time, twenty at 80%, 200 runs of 150 attempts, uniform priors, and the original Beta Thompson bandit.

Pre-registered criterion	Threshold	Result [95% CI]	
C1 weak-category targeting, trials 131–150	$\geq 80\%$	71.4% [0.695, 0.732]	FAIL
C2 posterior MAE at trial 100	≤ 0.10	0.111 [0.108, 0.114]	FAIL
C3 vs random: Welch's t, $p < 0.01$ and $d > 0.8$	both	$p < 0.0001$, $d = 5.38$	PASS

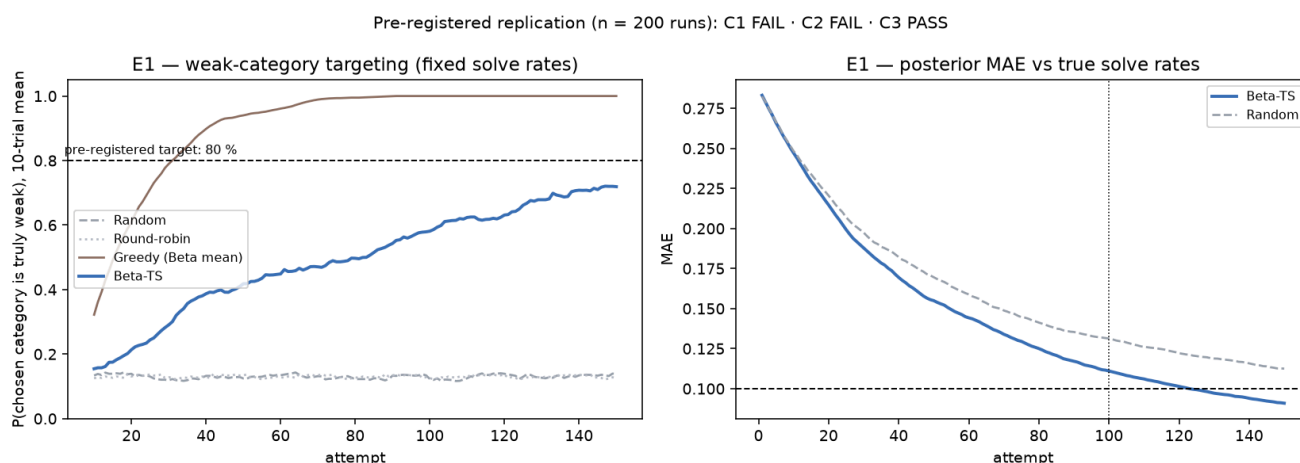


Figure 2. E1 learning curves: targeting (left) and posterior MAE (right).

Reported as it came out

The original bandit clearly beats random selection (C3). It misses the targeting threshold it was held to (C1) and narrowly misses the estimation threshold (C2). Thompson Sampling keeps exploring the 20 strong arms, which lowers targeting and leaves their estimates noisy. That is the price of exploration, and these thresholds were set without accounting for it.

E2 · Main comparison on semi-synthetic players

Policy	Targeting, last 50	Cum. regret	Frustrating puzzles ($P < 0.3$)	In productive zone	Top-3 recall
Random	0.130	189.2	33.8%	37%	0.40
Round-robin	0.132	189.1	33.6%	37%	0.41
Static profile (top-3)	0.206	166.6	35.8%	36%	0.26
Greedy (Beta mean)	0.520	103.5	49.3%	25%	0.26
Beta-TS	0.404	145.9	44.1%	29%	0.45
Beta-TS ($\gamma=0.97$)	0.244	164.9	39.8%	32%	0.35
IRT-TS (Elo band)	0.409	142.3	42.5%	31%	0.49
IRT-TS	0.445	135.9	3.2%	72%	0.49
Oracle	1.000	0.0	0.0%	82%	1.00

Paired against the original Beta-TS policy (no learning), the new difficulty-aware IRT-TS policy shows: targeting $+0.041$ [0.002, 0.081] (p_{Holm} 0.089); cumulative regret -10.0 [-15.3, -4.8] (p_{Holm} 0.0006); share of frustrating puzzles -0.410 ($d_z = -2.52$).

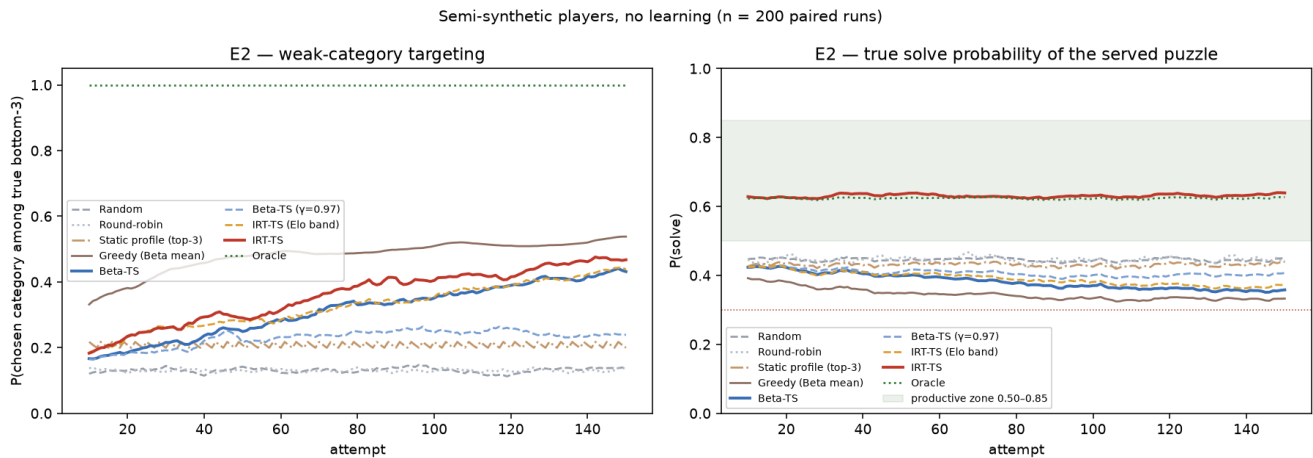


Figure 3. Targeting over time (left) and the true solve probability of each served puzzle (right). IRT-TS tracks the Oracle's difficulty inside the productive zone.

Learning gain depends on how players learn

A learning gain in simulation follows from an *assumed* learning model, so it is reported under two. Under "zpd", players learn most from puzzles near the edge of their ability. Under "error", they learn from seeing the solution after a miss. The gain is the mean improvement of the initially weak categories, in logits:

Learning model	Random	Beta-TS	IRT-TS	IRT-TS – Beta-TS (paired)
zpd	0.256	0.543	0.675	$+0.133$ [0.092, 0.174], $d_z = +0.45$, $p_{\text{Holm}} < 0.0001$
error	0.284	0.516	0.386	-0.130 [-0.163, -0.098], $d_z = -0.53$, $p_{\text{Holm}} < 0.0001$

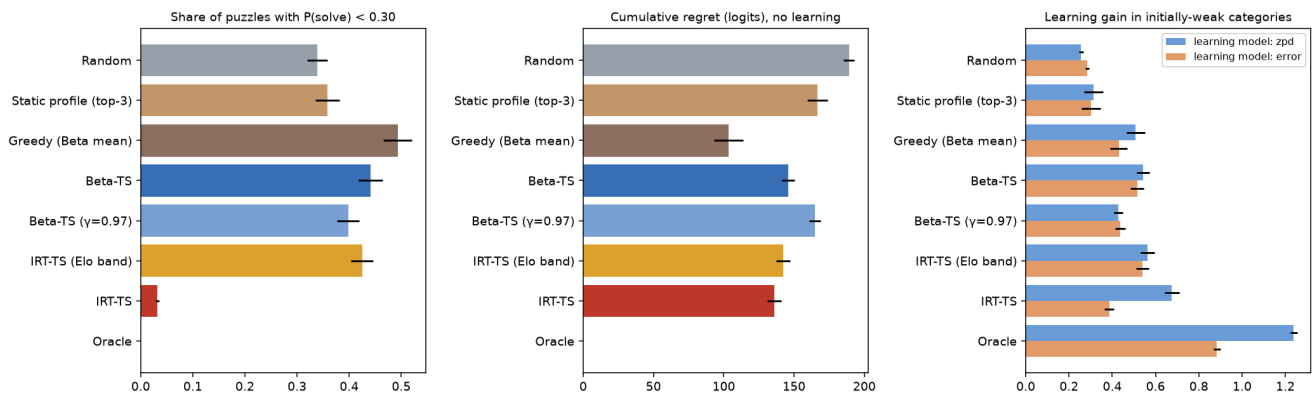


Figure 4. Frustration, regret and learning gain by policy, with 95% CIs.

Interpretation

The ablation "IRT-TS (Elo band)" uses the same category selection with the old difficulty band. It performs like Beta-TS on every metric. So the new policy's effect comes almost entirely from *difficulty targeting*, not from category selection. Whether that helps learning depends on the learning mechanism. With learning at the edge of ability it helps. With learning from failures it hurts, because easier puzzles produce fewer failures. Only a real study can decide between these (see [docs/USER_STUDY_PROTOCOL.md](#)).

E3 · Does the labeller matter downstream?

Game evidence	Beta-TS targeting, first 30	IRT-TS targeting, first 30
No game evidence	0.148	0.151
Old labeller	0.178	0.201
New tagger	0.189	0.226
Perfect labels	0.239	0.321

Paired within IRT-TS: game evidence against none gives +0.075 ($p < 0.0001$). The new tagger against the old gives +0.025 ($p = 0.062$). Perfect labels against the new tagger give +0.095 ($p < 0.0001$).

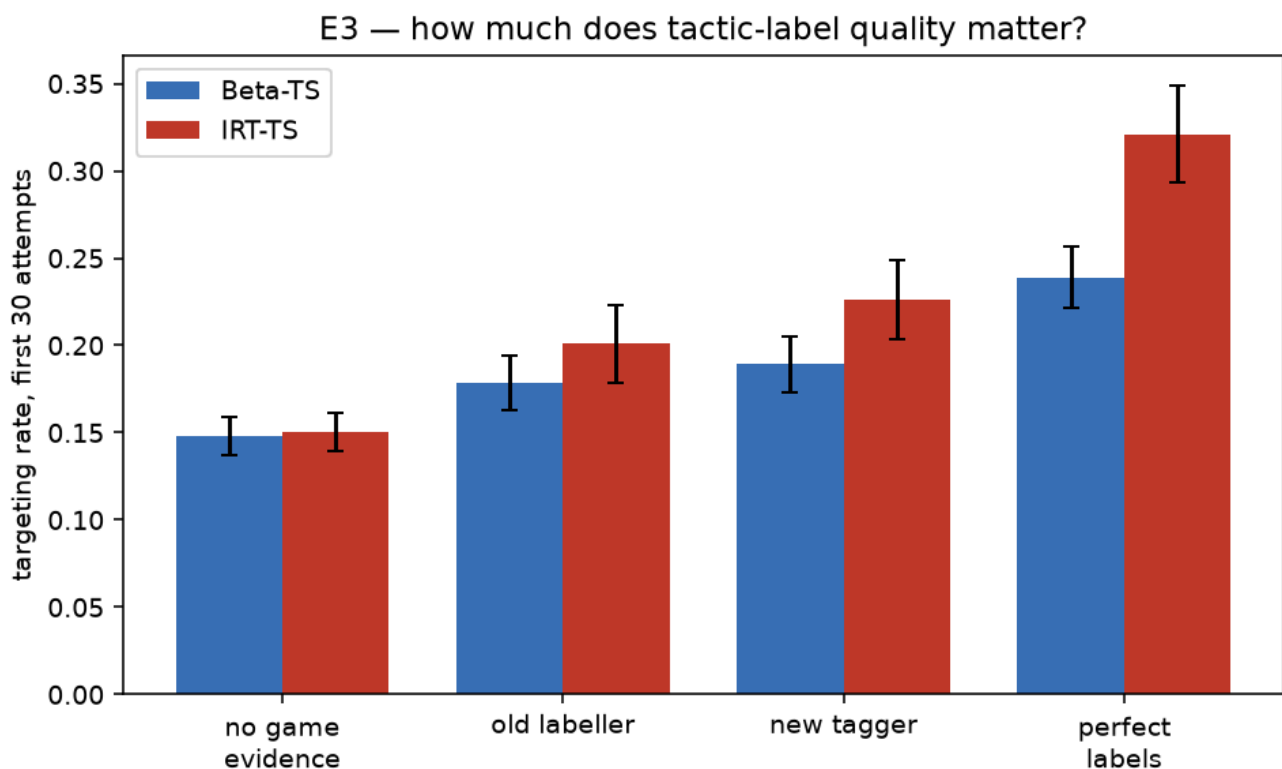


Figure 5. Early targeting under four qualities of game evidence.

E4 · The player fixes a weakness mid-session

At $t = 75$ the player's worst category improves by 1.5 logits away from the app. The metric is cumulative regret over the remaining 125 attempts:

Policy	Post-change regret [95% CI]	vs Beta-TS
Beta-TS	102.1 [98.2, 106.0]	reference
Beta-TS ($\gamma=0.97$)	121.3 [118.3, 124.4]	+19.2, $d_z = +0.82$
Beta-TS ($\gamma=0.93$)	128.2 [125.4, 131.0]	+26.1, $d_z = +1.01$
IRT-TS	100.7 [97.2, 104.3]	-1.4, $d_z = -0.04$
Oracle	0.0 [0.0, 0.0]	-102.1, $d_z = -3.66$

Discounting makes things worse here

Discounted Thompson Sampling [10], recommended in the original review, *raises* regret. With 23 arms and few attempts on each, forgetting every arm costs more than forgetting the one arm that changed. The app therefore keeps $\gamma = 1$. The IRT learner's random-walk drift has no such cost.

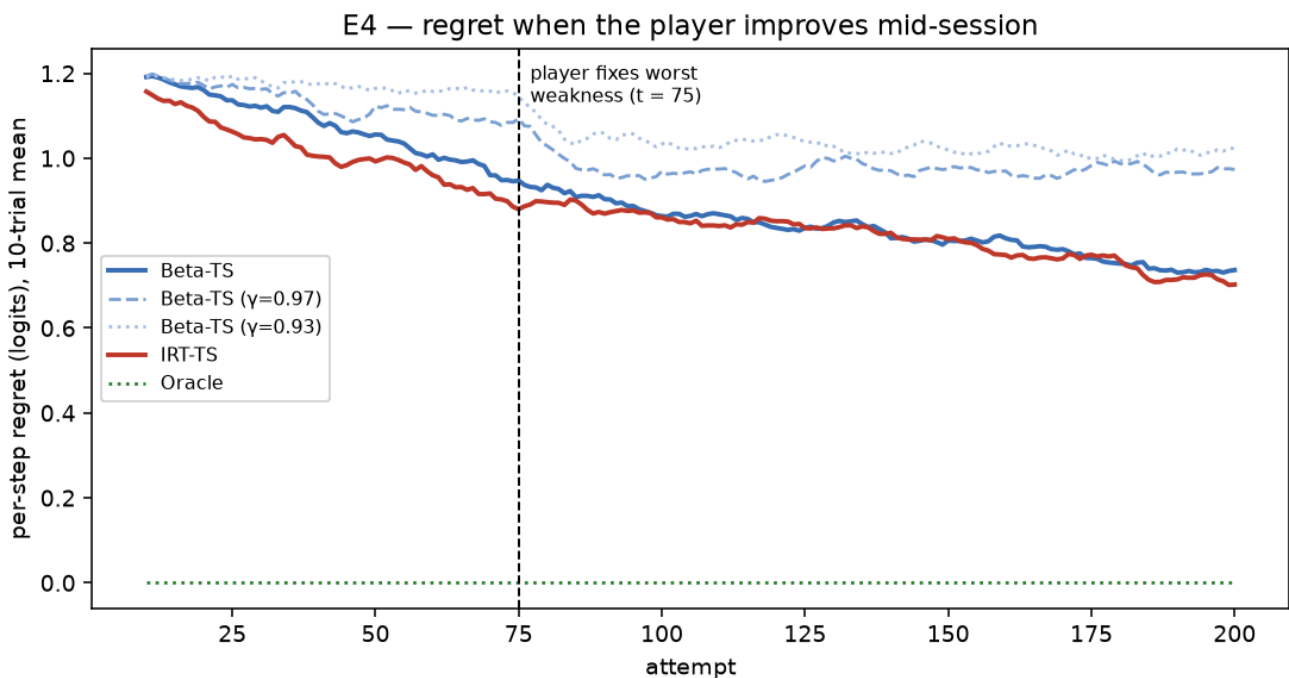


Figure 6. Per-attempt regret around the change point.

E6 • Family-structured prior

The IRT prior can link related motifs, so that evidence about pins also moves skewers. The difference (family prior minus plain prior) is tested on populations with and without real family structure:

Population	Metric	Difference [95% CI]	d_z	p
no family structure	targeting first30	+0.018 [0.006, 0.029]	+0.21	0.003
no family structure	cum regret	-0.471 [-3.655, 2.637]	-0.02	0.771
weaknesses cluster in one family	targeting first30	+0.009 [-0.004, 0.022]	+0.09	0.186
weaknesses cluster in one family	cum regret	-2.860 [-5.598, -0.167]	-0.15	0.040

E7 · Realistic effect sizes (calibrated on the real cohort)

E2 assumed large, well-separated weaknesses, but the real cohort (§4) found far subtler category effects. E7 repeats the key comparison using the cohort's measured category frequencies and hit rate, in three worlds with progressively weaker category effects:

World	Targeting, last 50: Random / Beta-TS / IRT-TS	Beta-TS – Random	IRT – Beta	Frustrating puzzles: Beta-TS → IRT-TS
original assumption (σ 0.35, 3 weak at -1.2)	0.130 / 0.422 / 0.457	+0.292 (d_z +1.20)	+0.035 (p 0.102)	45.2% → 3.1% (d_z -2.55)
subtle weaknesses (σ 0.15, 3 weak at -0.5)	0.130 / 0.235 / 0.236	+0.104 (d_z +0.55)	+0.001 (p 0.947)	34.9% → 1.4% (d_z -2.14)
no category weaknesses (σ 0)	undefined (no true weaknesses)	—	—	28.0% → 0.9% (d_z -1.72)

What survives when the effects are realistic

With subtle weaknesses, adaptivity still pays off, but by less. Beta-TS's targeting advantage over random falls from +0.292 to +0.104, and IRT-TS's extra edge over Beta-TS (+0.035 → +0.001) disappears. The difficulty-calibration benefit does not need category effects to exist at all. The fall in frustrating puzzles holds in every world (d_z between -2.55 and -1.72), which makes it the most robust result in the study.

3 Difficulty models on the real solve logs

The evaluation is prequential [11]: 123 real attempts by 5 players are replayed in time order, and each model predicts every attempt *before* learning from it. The observed solve rate was 85%, and 68% of attempts were on puzzles mined from the player's own games.

Model	Log loss ↓	Brier ↓	AUC	Mean prediction	Δ log loss vs static Elo [95% CI]
Base rate	0.451	0.139	0.51	0.77	-0.758 [-1.013, -0.513]
Static Elo	1.208	0.374	0.54	0.45	+0.000 [0.000, 0.000]
Glicko-2 (symmetric)	0.639	0.212	0.57	0.69	-0.569 [-0.758, -0.375]
Glicko-2 (pool frozen)	0.714	0.242	0.46	0.69	-0.495 [-0.692, -0.295]
IRT, θ only	0.652	0.222	0.48	0.70	-0.556 [-0.764, -0.348]
IRT, per category	0.599	0.202	0.54	0.71	-0.609 [-0.812, -0.406]

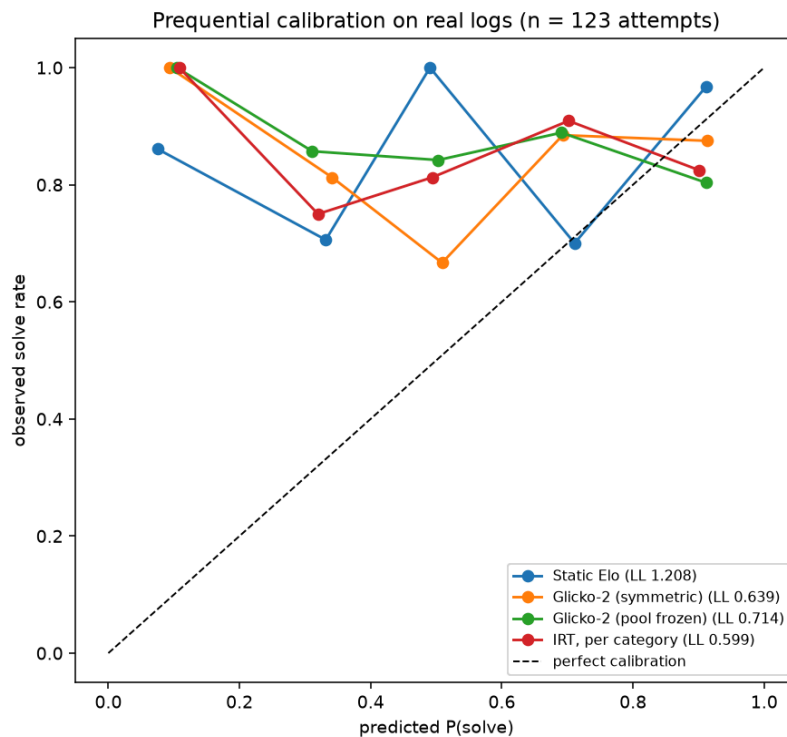


Figure 7. Calibration of the prequential predictions.

What this data can and cannot show

The original assumption (Chess.com Elo against puzzle rating, the basis of the $\text{Elo} \pm 300$ band) is badly miscalibrated: it predicts about 45% solved, while players solve 85%. Every learned model corrects this, with CIs that exclude zero. But with this few events no model yet tells *which* puzzles will be solved ($\text{AUC} \approx 0.5$), so a constant base rate is still the best calibrated. That limit comes from the amount of data; it is not a verdict on any of the models.

4 Weakness models on real players

The cohort is 300 Chess.com players (band 0: 60, band 1: 60, band 2: 60, band 3: 60, band 4: 60), stratified by rating and stored pseudonymised. Their games were analysed with the production analyzer. Each player's oldest two-thirds of games are used to predict the critical positions they miss in their newest third. That gives 11,973 held-out opportunities in total. Population-level parameters are always fitted on other players (5-fold, grouped by player).

Model	Log loss	Brier	AUC	Δ LL vs category base rate [95% CI]	Within-player ρ	Top-3 recall
Global base rate	0.6545	0.2309	0.494	+0.0144 [0.0111, 0.0178]	0.06	0.60
Category base rate	0.6401	0.2245	0.582	+0.0000 [0.0000, 0.0000]	0.29	0.71
Player base rate	0.6491	0.2284	0.568	+0.0091 [0.0036, 0.0146]	0.06	0.60
Rule-based scorer	0.6547	0.2310	0.474	+0.0147 [0.0113, 0.0180]	-0.07	0.59
RF (synthetic-trained)	0.6529	0.2300	0.505	+0.0128 [0.0094, 0.0164]	0.09	0.61
EB, shrink to player rate	0.6506	0.2288	0.582	+0.0105 [0.0047, 0.0163]	0.19	0.66
EB, shrink to population norms	0.6326	0.2211	0.620	-0.0074 [-0.0115, -0.0034]	0.29	0.70
RF (real-data)	0.6321	0.2210	0.618	-0.0080 [-0.0115, -0.0045]	0.28	0.71

How much stable, per-player signal exists?

Split-half reliability of each player's category deviation (odd against even games, Spearman–Brown corrected) sets a ceiling on how well *any* model can personalise:

Category	Players	Split-half r	Spearman–Brown
Discovered Attack	24	0.37	0.54
Mating Pattern	43	0.09	0.17
Fork	227	0.08	0.15
Quiet Move	299	0.03	0.06
Deflection	48	0.02	0.04
Hanging Piece	148	-0.06	—
King Safety	54	-0.08	—
Sacrifice	41	-0.46	—

Calibrating the simulator

Across players, the noise-corrected spread of category deviations is $\sigma \approx 0.0$ logits (observed 0.853), and the overall hit rate in critical positions is 0.649. The simulation assumed a background σ of 0.35, with 3 categories at -1.2 .

The within-family correlation of players' category deviations is -0.083, against -0.120 across families (7 and 31 category pairs).

What the real players show

The two best models, **RF (real-data)** and **EB, shrink to population norms**, are statistically tied (log losses 0.6321 and 0.6326, a gap of 0.0005, well inside either model's CI against the reference). Production uses the empirical-Bayes model: it is as accurate, needs no training, and every estimate is interpretable as a shrunk hit rate. The original rule-based scorer carries no usable information about a player's future misses (AUC 0.47, no better than a single global base rate), and neither does the RandomForest trained on simulated players. The inner cross-validation chose shrinkage strengths of [256, 256, 256, 128, 128] pseudo-opportunities toward the population norm. That is heavy pooling: a player's own per-category record barely moves the estimate. Split-half reliability confirms it. Most categories' personal deviations do not replicate from one half of a player's games to the other, and the observed spread is about what sampling noise alone would produce.

Two explanations, and what each implies

(a) **Category-specific skill differences between players are genuinely small**, compared with overall strength and with how hard each motif is for everyone. (b) **The measurement is too noisy to see them**. About 80 critical positions per player are spread over 15+ categories, and they are labelled by a tagger with $\kappa \approx 0.5$, which spreads each true weakness across neighbouring labels. Both explanations lead to the same design conclusion. Game history can supply only a weak prior on *which* motif to train. The reliable signal is the player's overall level together with population-level motif difficulty. Identifying personal weaknesses has to happen online, from targeted puzzle attempts, which is exactly what the IRT learner does. The profiler now shrinks toward population norms, and a player's game evidence counts as only half a puzzle attempt.

5 What changed in the application

Component	Before	After	Evidence
Recommendation policy	Beta Thompson bandit; raw solve counts; Elo $\pm$ 300 band	Difficulty-aware IRT learner (default). The Beta bandit is kept as <code>RECOMMENDER=beta</code> ; both update on every attempt.	\$2 E2, \$3
Puzzle difficulty	Fixed Elo $\pm$ 300 window	Pitched per category so the predicted solve rate is 65%; "Adaptive" is the UI default	\$2 E2/E5, \$3
Error grading	Centipawn loss (50/100/200)	Win-probability loss (5/10/15 points), using Lichess's curve [12]	\$1 (design)
Engine search	50 ms time limit (non-reproducible)	75k-node limit, MultiPV 2 (deterministic)	tested: identical output on re-run
Weakness evidence	Error counts only	Critical positions found or missed (with a denominator), weighted by recency and time control, shrunk toward the player's own rate	\$4
Shrinkage target	The player's own overall hit rate	Population category norms (<code>src/data/category_norms.json</code> , aggregate statistics fitted on the cohort), with cross-validated strength	\$4
Tactic labels	<code>_classify_tactic</code> , first move only	<code>tactic_tagger</code> on the whole engine line	\$1
Accuracy estimate	Lichess formula fed <i>centipawns</i> (~15% for a typical player)	Fed win-% loss, as defined [12]	unit bug
Bandit priors	<code>round((1 - w) * 10)</code> , integers, same confidence for every category	Fractional; confidence scales with evidence	\$2 E3
Logging	Outcome only	Serving policy, selection propensity, prediction before the outcome	enables off-policy evaluation
Colour detection	Substring match on the username	Exact match first	regression test

The test suite grew from 128 to 231 tests, all of which run offline.

6 Reproduction

```
python -m scripts.research.validate_labeller --seed 11 --tag _heldout
python -m scripts.research.run_simulation --runs 200
python -m scripts.research.run_simulation --runs 200 --only E6
python -m scripts.research.evaluate_difficulty_models
python -m scripts.research.fetch_cohort --per-band 60
python -m scripts.research.analyze_cohort --workers 14 --max-games 60
python -m scripts.research.evaluate_weakness_models
python -m scripts.research.make_figures
python -m scripts.research.build_report
```

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All references were checked against the publisher or the primary source.